

SLIC Methodology Technical Paper

English edition aligned with the live public board

Updated April 18, 2026. This compact paper is intended to keep the downloadable methodology artifact aligned with the live V3 public board.

SLIC is a public ranking instrument for smart and livable cities. The live board publishes one declared score per city, traces that score through five public pillars, and exposes the internal metric structure clearly enough for outside readers to audit the model.

This PDF focuses on the parts of the method that must remain stable across the site, exports, and published board: the final score equation, the disposable-income treatment, the normalization logic, the pillar weights, and the coverage logic.

1. Public Score Mechanics

The official published board uses one fixed weighted model. The five public pillars are aggregated once and only once in the final score.

$$SLIC(c) = 0.25 * Pressure(c) + 0.22 * Viability(c) + 0.18 * Capability(c) + 0.15 * Community(c) + 0.2$$

$$DI_ppp(c) = ((GrossIncome(c) * (1 - TaxRate(country(c)))) - Rent(c) - Utilities(c) - Transit(c)$$

$$s_m(c) = 100 * clamp((winsor(x_m(c)) - P5_m) / (P95_m - P5_m), 0, 1)$$

$$Coverage(c) = Sum\ observed\ gamma_m / Sum\ required\ gamma_m$$

Harmful indicators are reverse-scored inside the normalization step so that higher normalized values always indicate better urban outcomes. PPP conversion for disposable room is explicit and applied once inside the metric itself.

2. Public Pillar Weights

Pillar	Share of official score
Growth Pressure	25%
Viability	22%
Capability	18%
Community	15%
Creative	20%

3. Internal Metric Weights

The live public board currently uses the following metric weights inside each pillar. These values match the published ranking data export.

Pillar	Metric	Weight
Growth Pressure	Disposable income (PPP-adjusted residual room)	4
Growth Pressure	Housing burden	8
Growth Pressure	Economic growth momentum	6
Growth Pressure	Household debt burden	4
Growth Pressure	Working time pressure	7
Growth Pressure	Suicide / severe mental strain	7
Viability	Personal safety	5
Viability	Transit access and commute burden	5
Viability	Clean air	4
Viability	Water, sanitation, and utility reliability	4
Viability	Digital infrastructure	4
Viability	Climate and sunlight livability	7
Capability	Healthcare quality	8
Capability	Education quality	6
Capability	Equal opportunity and distributional fairness	4
Community	Hospitality and belonging	6
Community	Tolerance and pluralism	4
Community	Cultural and public-life vitality	7
Community	Birth rate optimism	2
Creative	Entrepreneurial dynamism	6
Creative	Innovation and research intensity	5
Creative	Economic vitality and productive context	5
Creative	Administrative and investment friction	4

4. Ranking Protocol and Publication Rules

- The public board uses one fixed weighted equation. Exploratory weight changes belong in the workbench, not in the official published score.
- Each published city shows one declared SLIC score, five public pillar values, and a visible coverage grade.
- City and metro data are preferred. When city data are thin, SLIC falls back in a visible order: subnational official data, national official data, then audited proxies.
- Satellite and remote-sensing layers are supporting evidence for air, ecology, and urban-edge context when ground sensors are sparse or spatially uneven.
- Ties are allowed. Cities with the same rounded final score can share the same numerical rank.
- Low-coverage cities should remain visibly labeled rather than being hidden behind false precision.

5. Interpretation Notes

- The published board is the baseline public board. The interactive workbench may rerank the broader indexed field, but it does not overwrite the published official score.
- Coverage grades matter. A city with a visible low-coverage label should be interpreted more cautiously than a city with dense, verified city-level evidence.
- Satellite evidence, open proxies, and analyst review can widen spatial coverage, but they should remain visible rather than being passed off as direct city-official measurements.

6. Selected Official and Public Sources

- 1 World Bank indicators and PPP series
- 2 World Health Organization (GHO OData API)
- 3 International Labour Organization (ILOSTAT)
- 4 WHO/UNICEF Joint Monitoring Programme for water and sanitation
- 5 UNESCO Institute for Statistics
- 6 OECD housing, health, and education datasets
- 7 OpenAQ and Copernicus / CAMS for air-quality context
- 8 Measurement Lab for digital performance context
- 9 World Intellectual Property Organization for innovation signals
- 10 World Meteorological Organization climate normals

7. Alignment Note

This PDF is deliberately compact. The website methodology page remains the fuller annotated reference, but this downloadable paper is intended to stay synchronized with the live formulas, weights, and export structure so readers do not encounter contradictory public documentation.